

SIMATS ENGINEERING

CO4 – AT1 - Article Review- Low

**COURSE NAME: Advance Wireless Communication Systems /
5G / 6G Communication Systems**
COURSE CODE: ECA56

COURSE OUTCOME COVERED

CO: 5 - *Develop 5G-enabled edge computing and IoT solutions for smart city, healthcare, industrial automation, and intelligent communication applications. (BL6)*

Assessment Tool Weightage: 25%

Short description about assessment: Students review recent technical articles to assess critical thinking and awareness of trends.

Dr.

QUESTIONS: Total Marks: Duration: 90 Minutes

1. Multi-access Edge Computing (MEC) Architectures for Low-Latency 5G Applications (BL5)
2. Role of MEC in Ultra-Reliable Low-Latency Communications (URLLC) (BL5)
3. Cloud Radio Access Network (Cloud RAN): Architecture, Benefits, and Challenges (BL5)
4. Virtualization and Resource Management in Cloud RAN (BL5)
5. Edge AI for Intelligent 5G Network Optimization (BL5)
6. Artificial Intelligence-Based Resource Allocation in Edge Computing (BL5)
7. Integration of MEC and Artificial Intelligence for Next-Generation Networks (BL5)
8. Security and Privacy Challenges in Edge Computing for 5G (BL5)
9. Dynamic Spectrum Sharing Techniques for Efficient Spectrum Utilization (BL5)
10. AI-Driven Dynamic Spectrum Sharing in 5G and Beyond Networks (BL5)
11. Satellite Backhaul Solutions for Rural and Remote 5G Connectivity (BL5)
12. Low Earth Orbit (LEO) Satellite Networks for 5G and 6G Communications (BL5)
13. Comparative Analysis of LEO, MEO, and GEO Satellites for Wireless Communication (BL4)
14. Integration of Terrestrial and Non-Terrestrial Networks (NTN) in 6G (BL5)
15. Edge Computing over Satellite Networks for Future Wireless Systems (BL5)

16. NB-IoT Architecture and Emerging Industrial Applications (BL5)
17. LTE-M Technology for Massive Machine-Type Communications (BL5)
18. Comparative Study of NB-IoT and LTE-M for IoT Deployments (BL4)
19. Power-Efficient Communication Techniques for IoT Devices in 5G Networks (BL5)
20. Battery Life Optimization Strategies for Massive IoT Devices (BL5)
21. Wearable Healthcare Devices Enabled by 5G Communication (BL5)
22. Smart Sensors for Intelligent IoT Applications in 5G Networks (BL5)
23. Energy-Efficient Communication Protocols for Large-Scale IoT Systems (BL5)
24. Intelligent Traffic Management Using 5G and Edge Computing (BL5)
25. AI-Enabled Smart Surveillance Systems over 5G Networks (BL5)
26. Smart Energy Monitoring and Management Using IoT and 5G (BL5)
27. Digital Twin Technology for Smart City Infrastructure (BL5)
28. Edge AI Applications for Sustainable Smart Cities (BL5)
29. Industrial IoT Applications in Smart Manufacturing (BL5)
30. Predictive Maintenance Using Artificial Intelligence and 5G Networks (BL5)
31. Robotic Automation in Industry 4.0 Using 5G Communication (BL5)
32. Private 5G Networks for Smart Factory Deployments (BL5)
33. Edge Computing for Real-Time Industrial Automation (BL5)
34. Remote Robotic Surgery Enabled by 5G Communication Networks (BL5)
35. AI-Driven Medical Diagnosis Using 5G and Edge Computing (BL5)
36. Intelligent Remote Patient Monitoring Using Wearable IoT Devices (BL5)
37. Secure Healthcare IoT Systems in 5G Networks: Challenges and Solutions (BL5)
38. Smart Irrigation Systems Using IoT and 5G Technologies (BL5)
39. Precision Agriculture for Crop Monitoring and Yield Prediction Using AI and 5G (BL5)
40. Intelligent Livestock Tracking and Smart Farming Using 5G IoT Networks (BL5)

ONE PAGE WRITEUP

Edge Computing for Real-Time Industrial Automation

Introduction:

Edge computing is an important technology for modern industrial automation because it brings data processing and decision-making closer to machines and sensors. Traditional industrial systems often send large amounts of data to a centralized cloud server, which can introduce communication delays and depend heavily on network connectivity. Edge computing reduces this delay by processing critical data locally at the factory or near the industrial equipment.

Working Principle

In an edge-based industrial automation system, sensors continuously collect information such as temperature, pressure, vibration, speed, current, and machine status. Edge devices such as industrial gateways, embedded controllers, or edge servers process this data in real time. Based on the analysis, the system can immediately control actuators, trigger alarms, adjust machine parameters, or stop equipment when an unsafe condition is detected. Only important or summarized information may be sent to the cloud for long-term storage and further analysis.

Applications

Edge computing can be used for predictive maintenance, robotic control, quality inspection, process monitoring, energy management, and worker safety. For example, vibration sensors can detect unusual machine behavior at the edge and identify possible equipment failure before it causes production downtime. In automated quality inspection, edge devices can analyze camera or sensor data quickly and reject defective products without waiting for a remote cloud response.

Advantages

The major benefits of edge computing are low latency, faster decision-making, reduced bandwidth usage, improved reliability, and better data privacy. Since critical processing occurs locally, industrial operations can continue even when the connection to the cloud is temporarily unavailable. Edge computing also supports scalable industrial IoT systems by distributing computational workloads across machines, gateways, and local servers.

Conclusion

Edge computing provides a practical foundation for real-time industrial automation by combining local data processing with intelligent control. It enables industries to respond quickly to changing machine conditions, improve productivity, reduce downtime, and increase operational safety. With the integration of Industrial IoT, artificial intelligence, and high-speed communication networks, edge computing is expected to play a major role in the development of smart factories and Industry 4.0.



Edge Computing for Real-Time Decision-Making in Industrial Automation Systems

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Abstract—Edge computing has emerged as a transformative paradigm for industrial automation, enabling real-time data processing and decision-making at the network's periphery. As industrial systems grow increasingly complex and latency-sensitive, traditional cloud-based approaches often fall short due to network congestion, high response times, and security concerns. This paper investigates the application of edge computing in real-time decision-making for industrial automation systems. We propose a layered architecture integrating edge nodes, intelligent gateways, and cloud backends to optimize operational efficiency, reduce latency, and enhance system resilience. A case study on a smart manufacturing line demonstrates up to 68% reduction in processing latency and improved fault detection accuracy. The findings underline the necessity of distributed intelligence and localized computation in achieving Industry 4.0 objectives.

Index Terms—Edge computing, real-time decision-making, industrial automation, cyber-physical systems, smart manufacturing, latency reduction, intelligent edge nodes, Industry 4.0.

Real-time decision-making in industrial automation systems demands not only fast computation but also context-aware processing and adaptive control mechanisms. Edge computing supports these requirements by combining computational intelligence with proximity to the physical processes. Through edge-enabled architectures, manufacturers can achieve faster event response, reduced data transmission, and improved privacy since sensitive data can be processed locally.

Despite its potential, the implementation of edge computing in industrial automation raises several challenges including architectural complexity, interoperability with legacy systems, device management, and security. Addressing these challenges requires a carefully designed architecture that harmonizes edge, fog, and cloud layers while leveraging modern networking protocols and AI-driven analytics.

This paper aims to explore the intersection of edge computing and real-time decision-making within the domain of industrial automation. We propose a multi-tiered system architecture that integrates intelligent edge devices, real-time data pipelines, and

Code of Conduct:

I P.HEMANTH REDDY (192312173) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P.Hemant reddy

RUBRICS FOR CALCULATION:

Article Review					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand